Material-sparing screening of amorphous solid dispersions using automated Raman spectroscopy

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Abstract

Automated experiments are playing a role in drug formulation development, including the design of amorphous solid dispersions (ASDs). ASDs, comprising a drug in the amorphous state dispersed in a polymer, are a formulation strategy to increase the solubility and thus bioavailability of poorly water-soluble drugs. In the early stages of developing ASDs, screening experiments should ideally be automated and material-sparing. The techniques used to analyse the samples must be sensitive to crystallinity and polymorphism and capable of analysing small sample volumes. Therefore, the capabilities and limitations of collecting and then analysing data from automated Raman spectroscopy were explored using Principal Component Analysis (PCA). Three plates at different drug-polymer concentrations were prepared and Raman spectra were automatically collected for each plate. PCA identified that samples on the 30 mg/mL plate were most suitable for automated data collection and it also highlighted that the reliability of automated Raman spectroscopy is dependent on whether the sample is dispensed centrally in the well. If these limitations are taken into consideration, automated Raman spectroscopy can be a reliable analysis technique in material-sparing screening experiments.

Introduction

Predictive methods using machine learning are increasingly playing a role in drug formulation development (1). These predictive models rely upon high-quality data to make meaningful predictions, and thus automated experiments have emerged to address the challenges around collecting data manually, such as irreproducibility, imprecision, and low throughput (2). Automation has been leveraged across many areas of drug formulation design. For example, automated experiments have been used to estimate kinetic parameters for secondary nucleation and crystal growth (3). They have been used to generate data to calibrate a predictive model of drug solubility in mixtures of organic solvents (4). They have also been used to determine the maximum concentrations of various drugs in amorphous solid dispersions (ASDs) (5). ASDs are a formulation strategy that is used to increase the solubility of poorly water-soluble drugs and consists of a drug in an amorphous state dispersed in a polymer (6). The physical stability of ASDs is one of the main challenges that must be overcome to increase the absorption of the drug in vivo (7). To do this, automation is now being employed to predict the stability of ASD formulations (8).

Crystalline solids and amorphous materials

When a solid is crystalline, its molecules are packed together in an ordered arrangement called a lattice. The lattice has long-range packing as it consists of a repeating three dimensional arrangement of molecules called the unit cell that extends throughout the solid (9). Lattices can be classified into 7 crystal systems by the shape of their unit cells and into 230 space groups by their symmetries (10).

The same crystalline solid can also exist in two or more distinct crystal structures called polymorphic forms. Each polymorphic form consists of a unique arrangement of the same molecules. As drugs typically exist as solids at room temperature and pressure, they also exhibit polymorphism. Importantly, different polymorphs of a drug can exhibit different physicochemical properties (11). The physicochemical properties of the drug influences its dissolution in the gastrointestinal tract, its absorption into the bloodstream, and thus its bioavailability, a key property of any oral formulation (12).

In contrast to crystalline solids, the molecules in amorphous materials are not packed together in an ordered arrangement. Amorphous materials do not consist of a lattice of repeating unit cells and thus they do not display long-range order. The disordered arrangement of its molecules means that amorphous materials have different physicochemical properties to the crystalline form of the same material (13). Most notably, amorphous materials have a higher free energy than their crystalline counterparts. This has the benefit that dissolution is a thermodynamically favourable process as they do not possess the lattice energy that must be overcome in crystalline solids. Amorphous materials therefore have faster dissolution rates and higher apparent solubilities (14). This is advantageous as a formulation strategy for poorly water-soluble drugs. These solubility benefits however must be balanced against the inherent thermodynamic instability of amorphous materials due to their higher free energy. This creates a thermodynamic drive towards crystallisation which is undesirable because the crystalline form typically exhibits lower solubility.

ASDs as a formulation strategy for poorly water-soluble drugs

Poorly water-soluble drugs are common in drug development pipelines. It has been reported that 70-90% of all new chemical entities have poor aqueous solubility (15). ASDs are a supersaturating drug delivery system designed to enhance the solubility of poorly water-soluble drugs. They dissolve the drug much more rapidly and at much higher concentrations than would be possible by dissolving the crystalline drug alone. Their function can be described by the ‘spring and parachute’ model. The amorphous drug acts like the ‘spring’ by dissolving rapidly and the polymer acts like the ‘parachute’, maintaining the metastable supersaturated concentration long enough for drug absorption to occur (16). The amorphous drug is also dispersed in a polymer to prevent

crystallisation which would hinder drug absorption (17). The polymer does this via thermodynamic and kinetic mechanisms. Thermodynamic mechanisms include the presence of drug-polymer interactions such as hydrogen bonding that can lower the free energy of the amorphous system and stabilise the amorphous drug. Kinetic mechanisms include a reduction in the mobility of drug molecules by the polymeric matrix, an increase in the glass transition temperature and an increase in the activation energy for crystallisation (18).

Characterisation of solid forms

Although crystallisation must be avoided in the manufacture of stable ASDs, during the preformulation phase of development, crystallisation is important for determining the drug loading capacity of the polymer (19). In these early stages of development, ASD screening experiments should be capable of differentiating between amorphous and crystalline samples and it would be useful if they could identify different polymorphic forms. Also the instrumentation used to screen ASD samples must be capable of analysing small sample volumes as at this stage experiments are conducted in a material-sparing manner to limit costs and emphasise sustainability.

X-ray diffraction is one such technique that is commonly used to characterise the solid form of a material (20). It can give information about the material's crystal structure such as lattice parameters and crystal symmetry (10). It can also distinguish between crystalline and amorphous materials as crystalline materials exhibit Bragg diffraction whereas amorphous materials do not. An X-ray diffractometer emits an X-ray beam at the sample and records the angle of reflection and the intensity of the elastically scattered X-rays. The X-rays reflect off of planes in the crystal lattice in accordance with Bragg's Law (equation (1)) which describes the relationship between the wavelength of the X-rays, λ, and the angle of reflection, θ, where d is the interplanar distance and n is an integer (21).

$$n\lambda = 2d\sin\theta \tag{1}$$

An alternative technique for studying crystal structures with a high sensitivity and selectivity for polymorphism is low frequency Raman spectroscopy (22). Raman spectroscopy involves directing a laser of a near-infrared wavelength (usually λ = 785 or 830 nm in most pharmaceutical applications) (23) at the sample and measuring the resulting wavelength of radiation that is scattered. The scattered radiation has a different wavelength to the incident radiation due to inelastic scattering caused by vibrations of the atoms in the solid. In the Raman spectrum produced, broad halos indicate amorphous material and sharp peaks indicate crystalline material with differences in wavenumber between peaks potentially indicating the presence of different polymorphs. Raman spectroscopy can be used in combination with X-ray diffraction (24, 25) but in this work, it was used alongside

polarised light microscopy like in other material-sparing screenings (26). Raman spectroscopy has numerous advantages. For example, spectra can be acquired much more rapidly with Raman spectroscopy compared to X-ray diffraction and in the context of process analytical technology, Raman spectroscopy can be used to analyse samples directly during manufacture whilst X-ray diffraction lacks this capability (22).

In this study, Raman spectroscopy was employed using an automated approach to analyse ASD samples which consisted of paracetamol dispersed in one of four different polymers. Paracetamol was chosen as the model drug because its physical properties and polymorphic forms are known and the polymers were chosen due to their pharmaceutical relevance and utilisation in similar works (7, 27). The results of Raman spectroscopy were compared to polarised light microscopy images which were used to visually assess the crystallinity of the samples. Principal Component Analysis was employed for dimensionality reduction and to identify trends in the dataset. The aim of this work was to explore the capabilities and limitations of analysing spectral data collected automatically using the Raman spectrometer across 5-2000 cm^{-1} and then focusing on the low-frequency (5-200 cm^{-1}) range.

Materials

Pharmaceutical grade paracetamol (PCM) was used as the drug in the ASD samples. *Affinisol™ 15LV* (AFF, Dow), hydroxypropyl methylcellulose acetate succinate HG (HPMCAS, Shin-Etsu), *Plasdone™ S630* (PLS) and *Soluplus®* (SOL, BASF) were used as the polymers in the ASD samples. Dimethyl sulfoxide (DMSO) was used as the solvent to dissolve the PCM and all four polymers during ASD sample preparation.

ASD sample preparation

PCM and the four polymers were prepared at three different concentrations, 60, 30 and 15 mg/mL respectively. The 60 mg/mL PCM stock solution was prepared by weighing 1500 mg PCM in a beaker, dissolving in DMSO and making up to 25 mL in a volumetric flask. A magnetic stirrer bar was added to the beaker and the solution was heated to 70 °C to fully dissolve the PCM in the DMSO. The initial PCM solution was then quantitatively transferred to the 25 mL volumetric flask. A 30 mg/mL PCM solution was prepared by extracting 12.5 mL of the PCM stock solution (equivalent to 750 mg PCM) and making up to 25 mL in a volumetric flask. A 15 mg/mL PCM solution was prepared by extracting 6.25 mL of the PCM stock (equivalent to 375 mg PCM) and making up to 25 mL in another volumetric flask. The AFF and PLS solutions were prepared following the same procedure as for the PCM solutions. HPMCAS and SOL were not soluble at 60 mg/mL, thus those solutions could not be prepared. The 30 mg/mL HPMCAS and SOL solutions were prepared by weighing 1500 mg of the polymer in a beaker, dissolving in DMSO and making up to 50 mL in a volumetric flask. The 15 mg/mL HPMCAS and SOL

solutions were prepared by extracting 25 mL of the solution (equivalent to 750 mg of the polymer) and making up to 50 mL in another volumetric flask.

To prepare the sample solutions, 200 µL mixtures of PCM and polymer were pipetted into glass vials at different drug loading concentrations according to Table 1.

Table 1

Volume of drug and polymer solutions in glass vials at each drug loading concentration.

Drug loading concentration (wt%)	Volume of PCM (µL)	Volume of polymer (µL)
100	200	0
95	190	10
90	180	20
85	170	30
80	160	40
75	150	50
70	140	60
0	0	200

The 100% drug loading was the drug control and the 0% drug loading was the polymer control. The sample solutions were not mixed since they contained two liquid mixtures in the same solvent so it was assumed that they would create a uniform mixture without additional agitation. 2 µL samples of the ASD mixtures were pipetted into the centre of each well of a 96-well glass plate. For each drug loading concentration, the ASD samples were pipetted in triplicate. Three plates were prepared with ASD mixtures consisting of PCM and the four polymers at equal solution concentrations, either 60, 30 or 15 mg/mL. In total, 288 ASD samples were prepared. Because HPMCAS and SOL were not soluble at 60 mg/mL, for the 60 mg/mL plate, 2 µL and 4 µL samples of PCM-AFF and PCM-PLS were pipetted but for the 30 and 15 mg/mL plates, only 2 µL samples were pipetted. The three plates were heated at 40 °C in a Thermo Scientific™ Vacutherm Vacuum Oven VT 6025 for 96 hours to evaporate the DMSO.

Data collection

Images of the three plates were captured using the Leica DM6000 M Automated Research Microscope (Figure 1). Full-size polarised light microscopy images of the three plates are provided in Appendix A. Data was

collected on the three plates using the HORIBA LabRAM Odyssey Raman spectrometer. Spectra were obtained automatically for all the samples on the three plates using the MultiWell module in HORIBA's LabSpec 6 Spectroscopy Suite software platform. A glass reference spectrum was obtained manually using the same instrument.

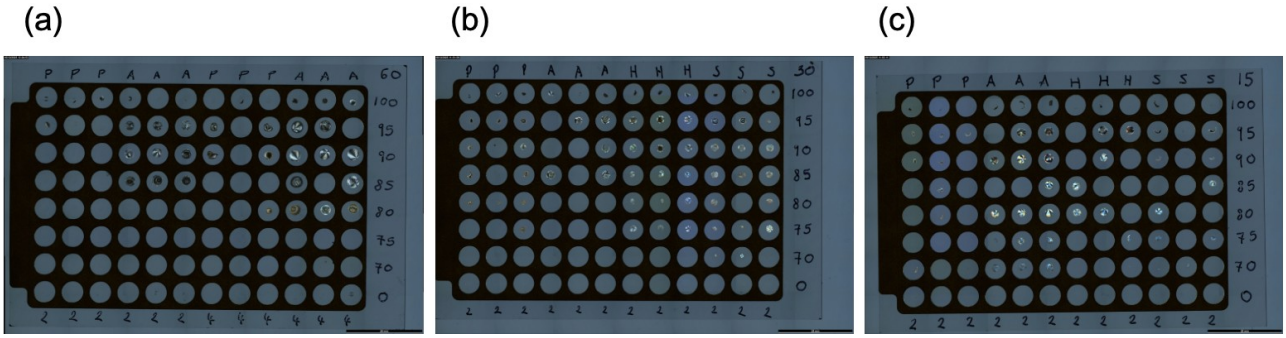

Figure 1. Polarised light microscope images of **(a)** the 60 mg/mL plate, **(b)** the 30 mg/mL plate and **(c)** the 15 mg/mL plate.

Data analysis

Preprocessing of the dataset was conducted for the purpose of removing unwanted or uncontrolled effects (28). Firstly, standard normal variate (SNV) was applied to normalise each spectrum to a mean of zero and a standard deviation of one. This was done to reduce any offsets in the spectra which can arise from variations in sample thickness and lead to differences in Raman signal intensity. Then the Savitzky-Golay (SG) filter with first derivative was applied to smooth the spectra and display the rate of change of intensity with respect to wavenumber. Following this, Principal Component Analysis (PCA) was conducted with three principal components. Three principal components were chosen as they explain 78.1% of the variance in the data when PCA was conducted from 5-2000 cm-1 and 89.4% of the variance when PCA was conducted from 5-200 cm-1 (Appendix B). For PCA in the 5-2000 cm-1 range, SG with derivative order 1, polynomial order 1 and window size 15 was applied. For PCA in the 5-200 cm-1 range, SG with derivative order 1, polynomial order 1 and window size 8 was applied. PCA was conducted in Python 3.11 using pyPhi v4.01 (29) and Raman spectra were produced using RamanSPy (30).

Results

PCA was firstly conducted on the entire dataset from 5-2000 cm^{-1} (Figure 2).

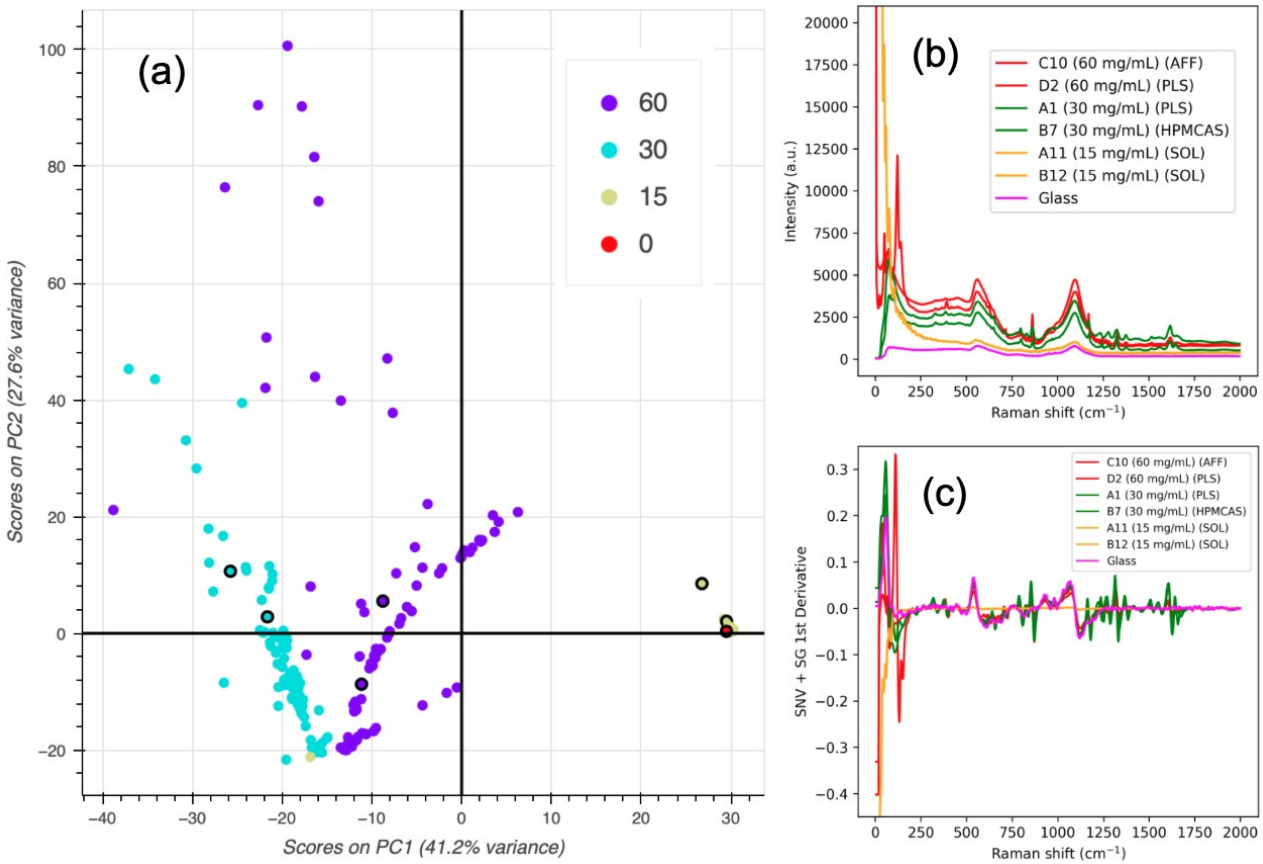

Figure 2. PCA was conducted from 5-2000 cm^{-1} for all samples across all three plates (including the glass reference sample). **(a)** Score scatter plot of PC1 vs PC2 coloured by concentration in mg/mL. The variance captured by each principal component is in brackets. 0 mg/mL refers to the glass reference sample. Samples circled in black are shown in (b) and (c). **(b)** Unprocessed Raman spectra from 5-2000 cm^{-1} displaying a subset of samples from each plate. **(c)** Preprocessed Raman spectra from 5-2000 cm^{-1} displaying a subset of samples from each plate.

When scores for the first two principal components are plotted, the glass reference sample and samples from the 15 mg/mL plate are separated from samples on the 60 mg/mL and 30 mg/mL plates by PC1 (Figure 2a). One of the samples in the 15 mg/mL plate does not follow this trend as its scores closely match those of samples on the higher concentration plates. There also appears to be some separation between the samples on the 60 mg/mL and 30 mg/mL plates, although there is a sample from the 60 mg/mL plate whose score on PC1 is more negative than all other samples at its concentration. PC2 largely captures the variance across the samples in the 60 mg/mL and 30 mg/mL plates and only separates two samples from the 15 mg/mL cluster. Plotting the raw spectra shows that the 15 mg/mL samples and the glass reference closely resemble each other in terms of shape with two broad peaks at *ca.* 600 and 1100 cm^{-1} (Figure 2b). Although the higher concentration plates also share those peaks, in addition, they have common sharper peaks at *ca.* 900, 1300 and 1600 cm^{-1}. The higher concentration plates also have higher Raman signal intensities than the 15 mg/mL plate and glass reference

from 250-2000 cm^{-1}. The preprocessed Raman spectra highlights the differences between the higher and lower concentration plates as there are distinct features in the higher concentration plates at *ca.* 900, 1300 and 1600 cm^{-1} that are absent in the lower concentration plates (Figure 2c). Changes in slope in the raw spectra are amplified in the preprocessed spectra making them much more visible to the naked eye.

To explain the separation in the 60 mg/mL and 30 mg/mL samples by PC1 (Figure 2a), PCA was conducted a second time from 5-2000 cm^{-1} with the 15 mg/mL samples and the glass reference excluded from the input dataset provided to the model (Figure 3).

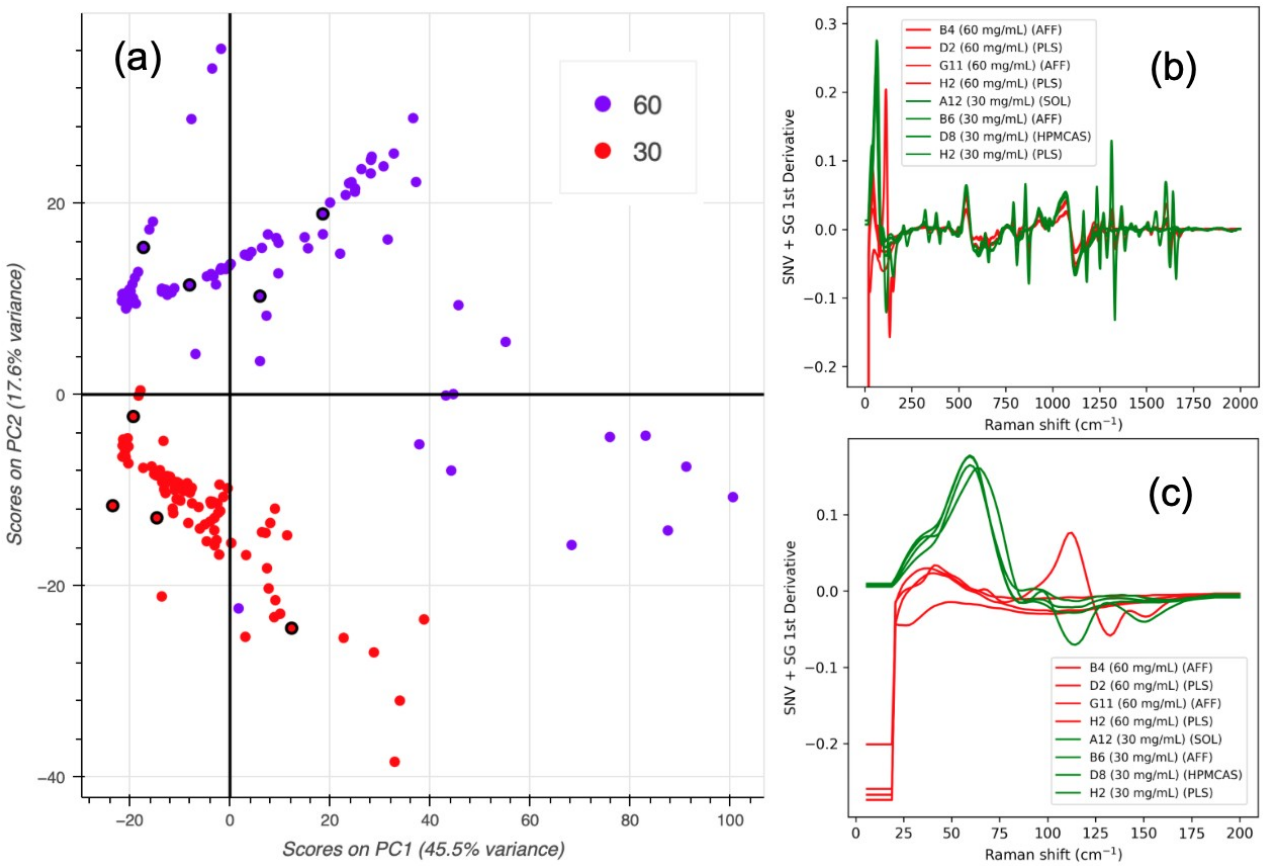

Figure 3. PCA was conducted from 5-2000 cm^{-1} but only for samples from the 60 mg/mL and 30 mg/mL plates. **(a)** Score scatter plot of PC1 vs PC2 coloured by concentration. The variance captured by each principal component is in brackets. Samples circled in black are shown in (b) and (c). **(b)** Preprocessed Raman spectra from 5-2000 cm^{-1} displaying a subset of samples from the 60 mg/mL and 30 mg/mL plates. **(c)** Preprocessed Raman spectra from 5-200 cm^{-1} displaying a subset of samples from the 60 mg/mL and 30 mg/mL plates.

Scores for the first two principal components were plotted but now PC2 appears to separate samples by concentration instead of PC1 (Figure 3a). 86 out of 96 samples from the 60 mg/mL plate (89.6%) have positive scores on PC2 whereas 95 out of 96 samples from the 30 mg/mL plate (99.0%) have negative scores on PC2.

Plotting the preprocessed Raman spectra for a subset of samples from each cluster, the 60 mg/mL and 30 mg/mL samples mostly overlap from 250-2000 cm^{-1} (Figure 3b) but in
Figure 3c, we see clear differences in the spectra in the low-frequency region (5-200 cm^{-1}). The 30 mg/mL samples have a broad peak at *ca.* 60 cm^{-1} whilst the 60 mg/mL samples do not. Therefore the low-frequency region is most sensitive to differences in these higher concentrations.

PC1 and PC2 from 5-2000 cm^{-1} have separated samples by concentration but there are other variables of interest that the first two principal components have not been able to separate the samples by (Appendix C). We have also not had to use PC3 as the first two principal components have explained on average 65.9% of the variance across Figures 2 and 3. Since the low-frequency region has a greater degree of specificity and sensitivity for differentiating between polymorphs (22), PCA was conducted a third time from 5-200 cm-1 to observe if the model could differentiate between crystalline and amorphous samples (Figure 4).

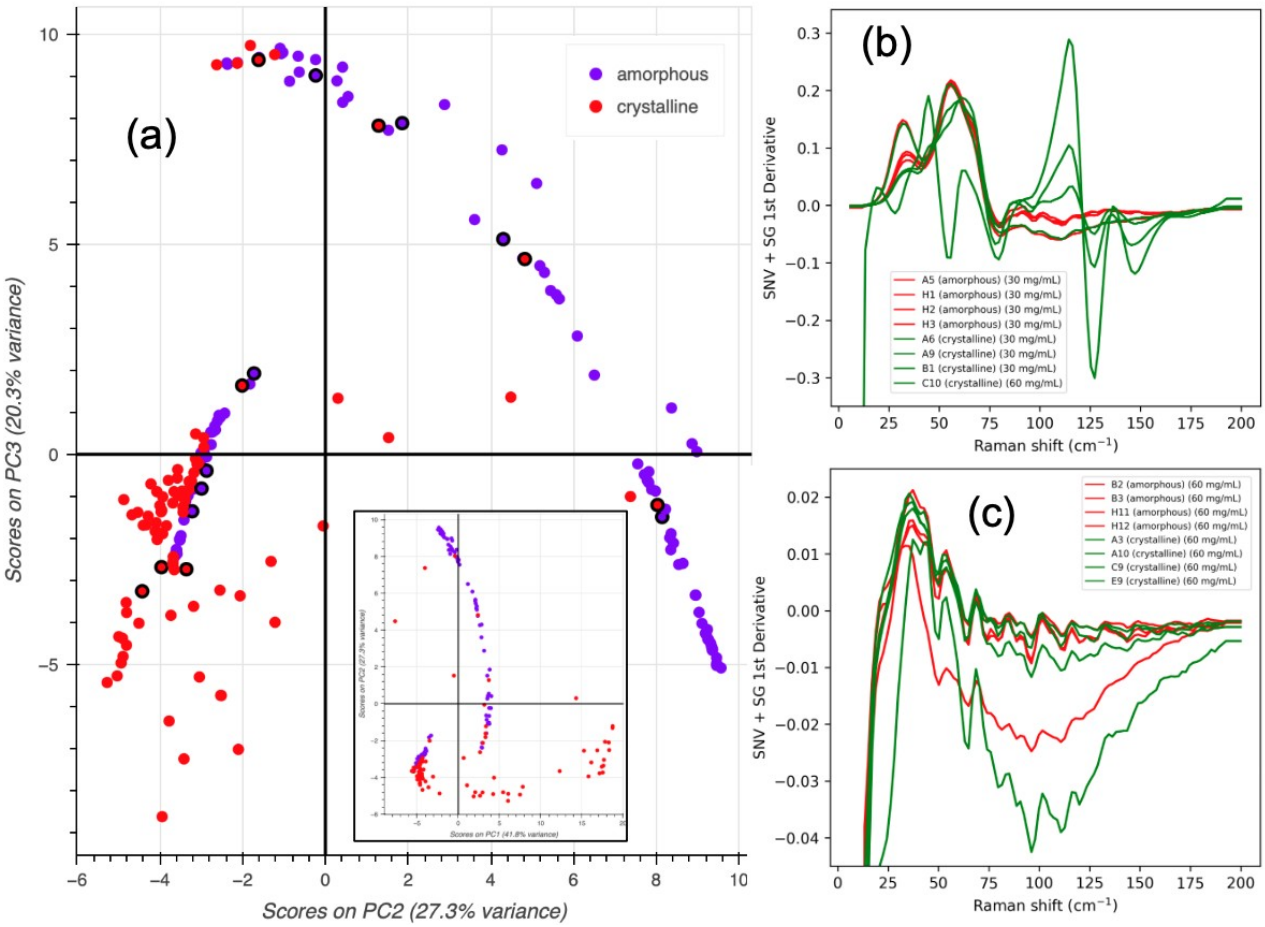

Figure 4. PCA was conducted from 5-200 cm^{-1} for all samples from the 60 mg/mL and 30 mg/mL plates. **(a)** Score scatter plot of PC2 vs PC3 coloured by whether the sample is crystalline or amorphous. The variance captured by each principal component is in brackets. The corresponding score scatter plot of PC1 vs PC2 is

inset in the bottom right quadrant for reference. Samples circled in black are shown in (b) and (c). The sample's crystallinity was visually assessed using polarised light microscopy. **(b)** Preprocessed Raman spectra from 5-200 cm^{-1}. A subset of amorphous samples are displayed in red. Crystalline samples in the crystalline cluster are displayed in green. **(c)** Preprocessed Raman spectra from 5-200 cm^{-1}. A subset of crystalline samples are displayed in green. Amorphous samples in the amorphous cluster are displayed in red.

When PC1 was plotted against PC2, some separation was observed (Figure 4a inset) but when the scores for PC2 were plotted against the scores for PC3, clearer separation between the amorphous and crystalline samples could be observed (Figure 4a). However, this separation is not entirely clear as 39 out of the 106 amorphous samples (36.8%) are in the crystalline cluster. Figure 4b highlights this discrepancy as it shows the preprocessed Raman spectra of a subset of the amorphous samples that are present in the crystalline cluster. The amorphous samples show an increase in the first derivative from 25-60 cm^{-1} followed by a plateau from 75-200 cm^{-1} whereas the crystalline samples have a change in the first derivative from positive to negative at *ca.* 120 cm^{-1} indicating a peak in the unprocessed Raman spectra. It is unclear why their scores are closely aligned on the score scatter.

9 out of the 86 crystalline samples (10.5%) are also present in the amorphous cluster. Figure 4c displays the preprocessed Raman spectra of a subset of the crystalline samples in the amorphous cluster. Both the crystalline and amorphous samples share similar spectral features from 5-120 cm^{-1}. It is however unclear why crystalline samples with such different appearances on the plate would have similar spectral features to amorphous samples, for example, samples C9, E9 and H11 from the 60 mg/mL plate (Appendix A: Figure 1).

Finally, for the purpose of assessing whether the PCA model would be useful for solid form characterisation, PCA was conducted for a final time from 5-200 cm^{-1} but only on the crystalline samples from the 60 mg/mL and 30 mg/mL plates (Figure 5).

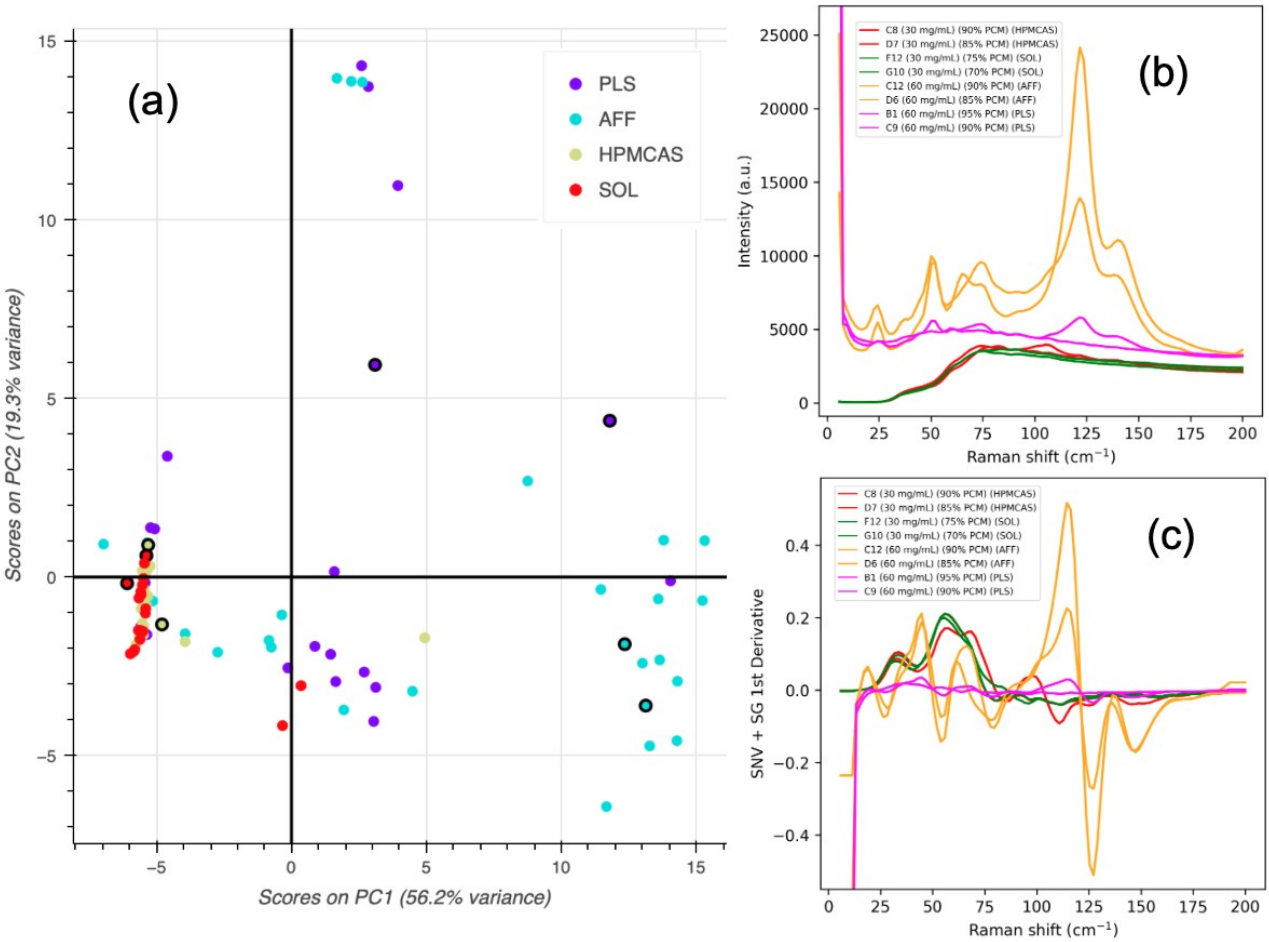

Figure 5. PCA was conducted from 5-200 cm^{-1} but for only the crystalline samples from the 60 mg/mL and 30 mg/mL plates. **(a)** Score scatter plot of PC1 vs PC2 coloured by polymer. The variance captured by each principal component is in brackets. Samples circled in black are shown in (b) and (c). **(b)** Unprocessed Raman spectra from 5-200 cm^{-1} displaying two samples for each polymer. **(c)** Preprocessed Raman spectra from 5-200 cm^{-1} displaying two samples for each polymer.

When the scores for PC1 were plotted against the scores for PC2 and coloured by polymer (Figure 5a), ASD samples containing SOL and HPMCAS are predominantly clustered together with negative PC1 scores and PC2 scores of ± 2. In contrast, AFF and PLS are highly dispersed along both principal components and do not form clusters. Comparing this to the Raman spectra in Figures 5b and 5c, we can observe that SOL and HPMCAS overlap tightly across the entire low-frequency range whereas there is much greater variability in the AFF and PLS spectra (although they also share common spectral features such as peaks at *ca.* 50 and 125 cm^{-1}).

Discussion

The PCA model seems to be able to separate samples by concentration with a high degree of accuracy as PC1 could separate the higher concentration samples from the lower concentration ones (Figure 2a) and PC2 could distinguish between the 60 mg/mL and 30 mg/mL samples (Figure 3a). The model could also distinguish between crystalline and amorphous samples although with less accuracy as amorphous samples appeared in the crystalline cluster and vice versa (Figure 4a). It should be noted that those discrepancies could be a result of the manual identification of whether a sample is crystalline or amorphous using polarised light microscopy, instead of a more objective technique like X-ray diffraction. Also, using polarised light microscopy, crystallinity is detected macroscopically by observing the whole well whereas Raman spectra were only collected at specified, microscopic locations in the well. These locations may not have been rich in crystalline material and may not be representative of the sample in the well.

The model also provided information regarding the quality of the data collected. The 15 mg/mL samples were tightly clustered around the glass reference (Figure 2a). This shows that there were minimal differences between those spectra and the glass reference meaning there was insufficient data quality in those samples for the PCA model to make distinctions between their scores. This might have been because there was too little ASD material in the well so the Raman signal intensity was very low and thus was overpowered by the glass background present in every sample.

In contrast, in the case of the 60 mg/mL plate, despite its separation from the 30 mg/mL plate by PC2 (Figure 3a), there was little difference between the two concentrations in the preprocessed spectra from 250-2000 cm^{-1} (Figure 3b) and they mostly differed in the low-frequency region. Doubling the concentration appeared to have little effect on the preprocessed Raman spectra, suggesting that the 30 mg/mL concentration of PCM and the four polymers might be sufficient for obtaining Raman spectra of a high enough intensity for the PCA model. In addition, 60 mg/mL exceeded the solubility of HPMCAS and SOL in DMSO. AFF was also highly viscous at that concentration and bubbles were produced during stirring which may have introduced errors during transfer and dilution. This begs the question: if PC1 is the direction of maximum variance in the data and in our model, it mostly captures the differences in concentration, if all the samples were at a concentration of 30 mg/mL, what variance would PC1 have captured? It is worth considering if PC1 may have captured differences between polymers or drug loadings and in the low-frequency region, perhaps differences between polymorphs. For example, even when conducting PCA on only the 30 mg/mL plate, as scores on PC1 became more negative, drug loadings tended to increase (Appendix D) although this correlation was not very strong.

Some of the shortfalls in the PCA model might also be explained by the limitations of the MultiWell module in HORIBA's LabSpec 6 Spectroscopy Suite. When collecting Raman spectra autonomously, spectra are collected at 1, 9 or 25 positions in the well, starting from the centre of the well and then forming a 3x3 or 5x5 grid. The resulting spectrum for each well is the average of the spectra collected at each position. The advantage of this approach is that data is collected rapidly and only one spectrum is produced per well. However, this approach can miss points of interest that are not in the centre of the well. To increase the probability that points

of interest are central in the well, an autonomous liquid handler could have been used to pipette the ASD samples into the wells. In addition, it may have been necessary to mix the sample solutions in the glass vials during ASD sample preparation to properly disperse the drug within the polymer. By not doing this, the ASD mixtures may not have been homogenous and thus regions of crystallinity could have appeared at the edges of the droplet pipetted into the centre of each well and been missed. It was assumed that since the drug and the polymer were dissolved in the same solvent, combining the two solutions would form a uniform mixture although it may have been more prudent to mix the sample solutions directly.

A further issue with the MultiWell module that would hinder its applicability in automated screenings is that currently it does not enable autofocus. This means that the quality of the data collected is dependent on the researcher manually adjusting the focus for each plate. This can delay data acquisition and is prone to user error.

Conclusion

PCA was a useful tool for analysing spectral data collected automatically using Raman spectroscopy. The 5-2000 cm^{-1} range showed that a sample concentration of 15 mg/mL is unsuitable for data collection using an automated approach. It also showed that a concentration of 60 mg/mL does not present a substantial improvement in data quality compared to the 30 mg/mL concentration. Therefore 30 mg/mL is the most suitable concentration for automated Raman spectroscopy out of the three that were tested. PCA also suggested that the MultiWell module in HORIBA's LabSpec 6 Spectroscopy Suite has certain limitations but that these can be mitigated by using an autonomous liquid handler and ensuring that the drug is uniformly dispersed in the polymer.

Data availability

The data analysis workflow can be accessed via the GitHub repository: https://github.com/lca6/PCM_ASDs_PCA. The full Raman spectra dataset is available in the "data" folder in the repository.

Author contributions

L. Morelli: Formal analysis, Investigation, Visualization, Writing - original draft

M. Devlin: Conceptualization, Methodology, Supervision, Writing - review & editing

M. Prostredny: Conceptualization, Software, Supervision, Writing - review & editing

AJ. Florence: Project administration, Supervision

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Appendix A

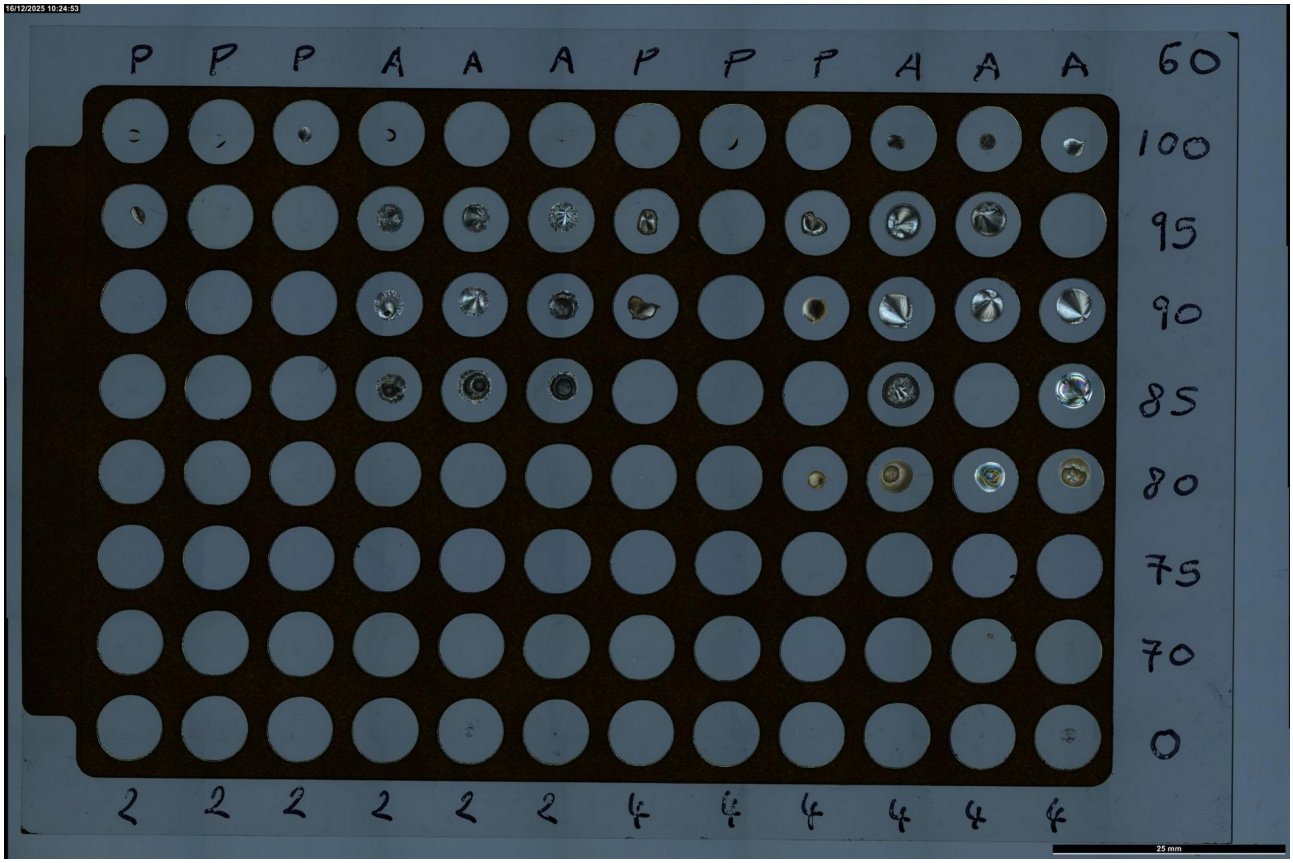

Figure 1. Polarised light microscope image of 60 mg/mL plate. The top column labels indicate the polymer used. “P” means the column contains PCM-PLS samples. “A” means the column contains PCM-AFF samples. The row labels indicate the drug loading concentration. The bottom column labels indicate the sample volume, either 2 or 4 μL.

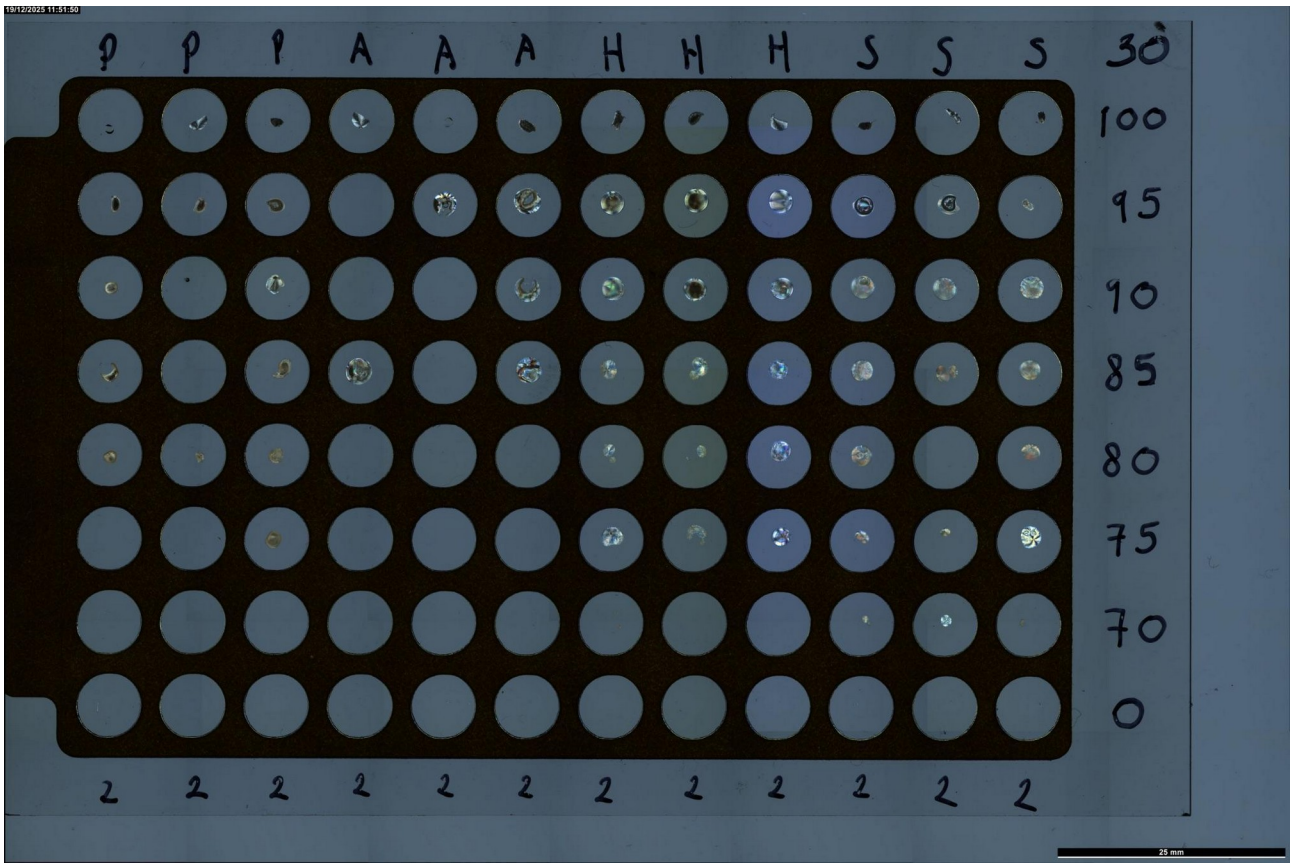

Figure 2. Polarised light microscope image of 30 mg/mL plate. The top column labels indicate the polymer used. “P” means the column contains PCM-PLS samples. “A” means the column contains PCM-AFF samples. “H” means the column contains PCM-HPMCAS samples. “S” means the column contains PCM-SOL samples. The row labels indicate the drug loading concentration. The bottom column labels indicate the 2 µL sample volume.

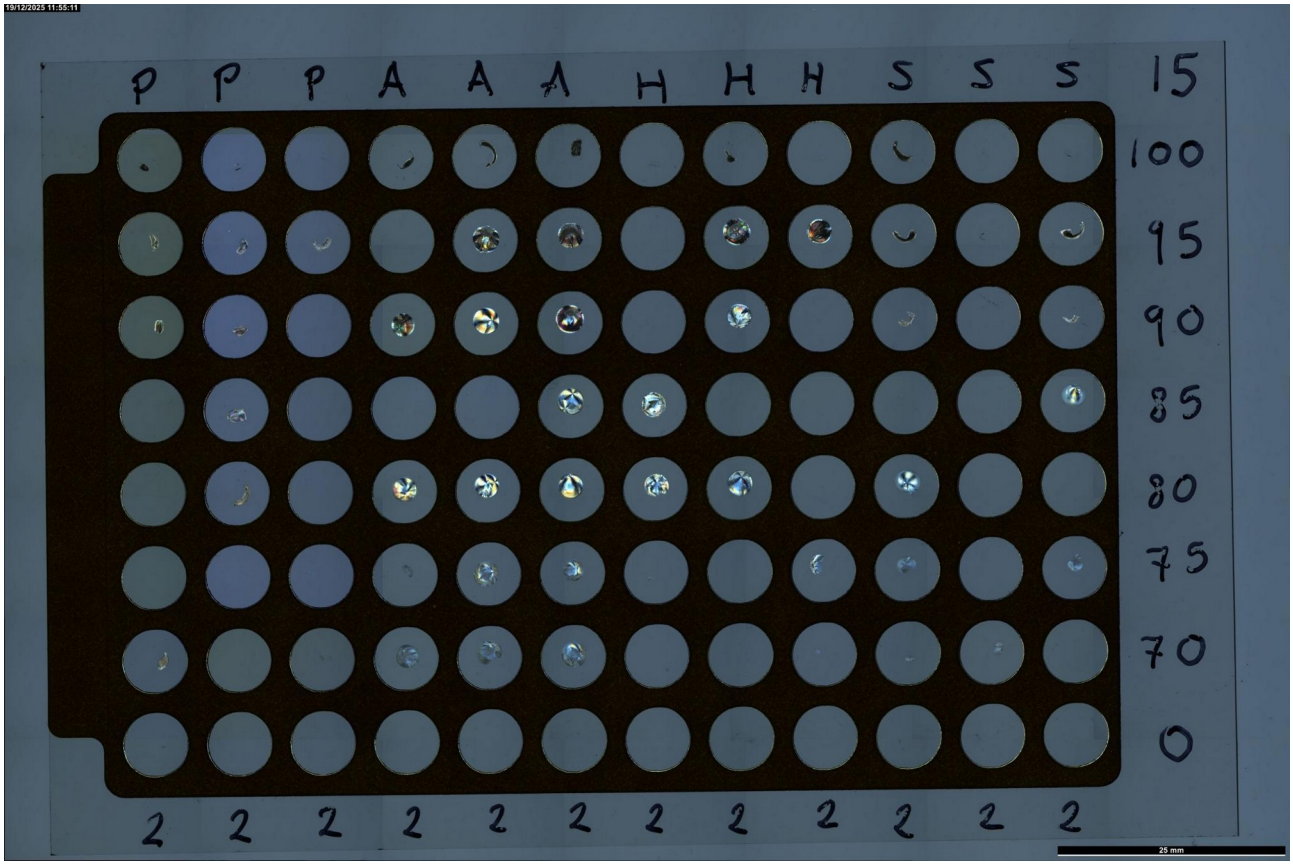

Figure 3. Polarised light microscope image of 15 mg/mL plate. The top column labels indicate the polymer used. “P” means the column contains PCM-PLS samples. “A” means the column contains PCM-AFF samples. “H” means the column contains PCM-HPMCAS samples. “S” means the column contains PCM-SOL samples. The row labels indicate the drug loading concentration. The bottom column labels indicate the 2 µL sample volume.

Appendix B

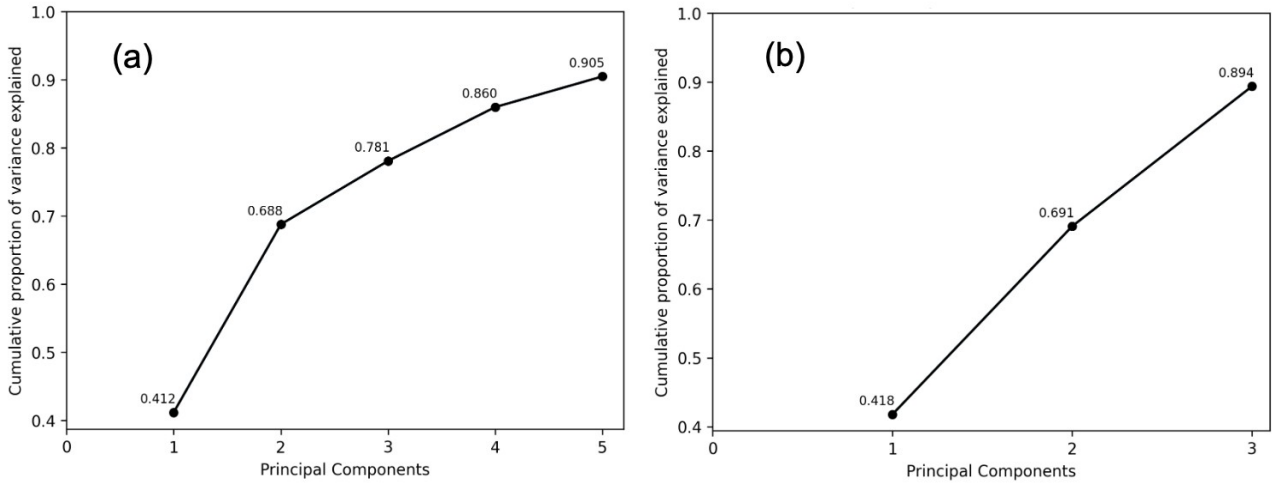

Figure 1. Line graphs displaying the number of principal components against the cumulative proportion of variance that they explain. **(a)** is for when PCA was conducted from 5-2000 cm^{-1} and **(b)** is for when PCA was conducted from 5-200 cm^{-1}.

Appendix C

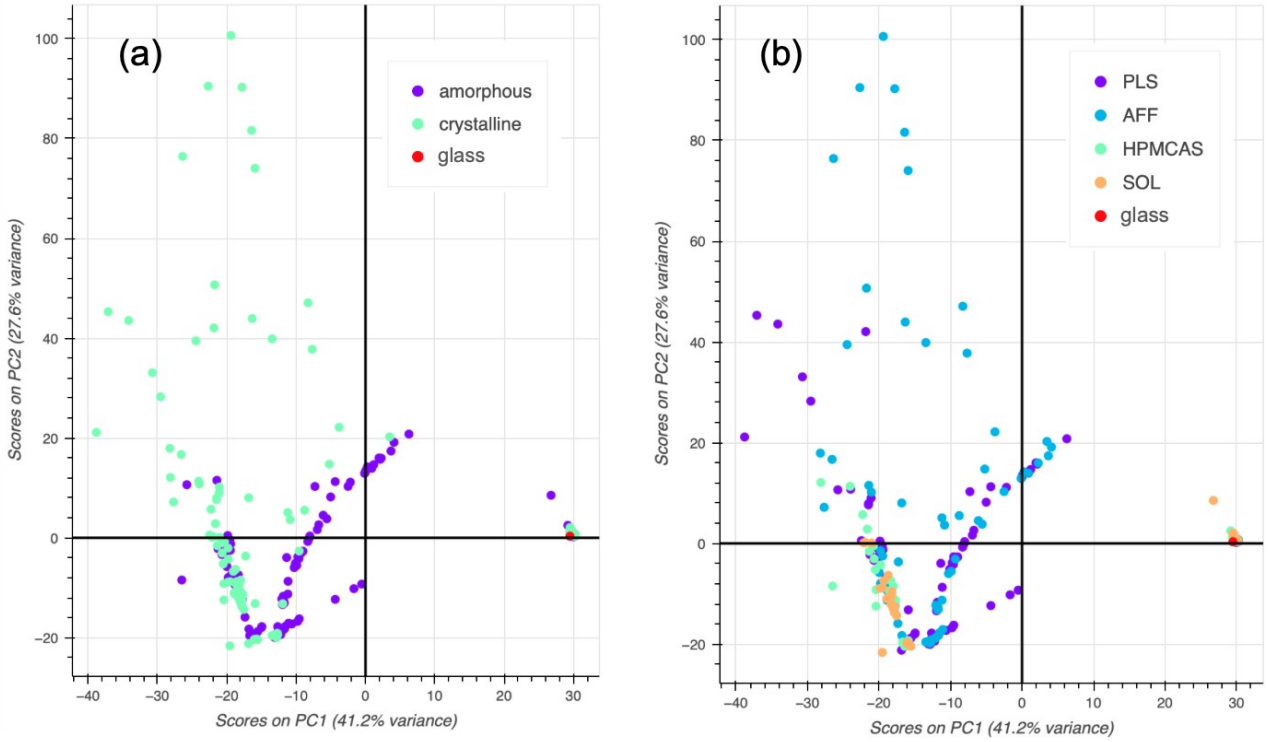

Figure 1. Score scatter plots of PC1 vs PC2 for all samples across all three plates (including the glass reference sample) coloured by **(a)** crystalline vs amorphous and **(b)** polymer. PCA was conducted from 5-2000 cm^{-1}. The variance captured by each principal component is in brackets.

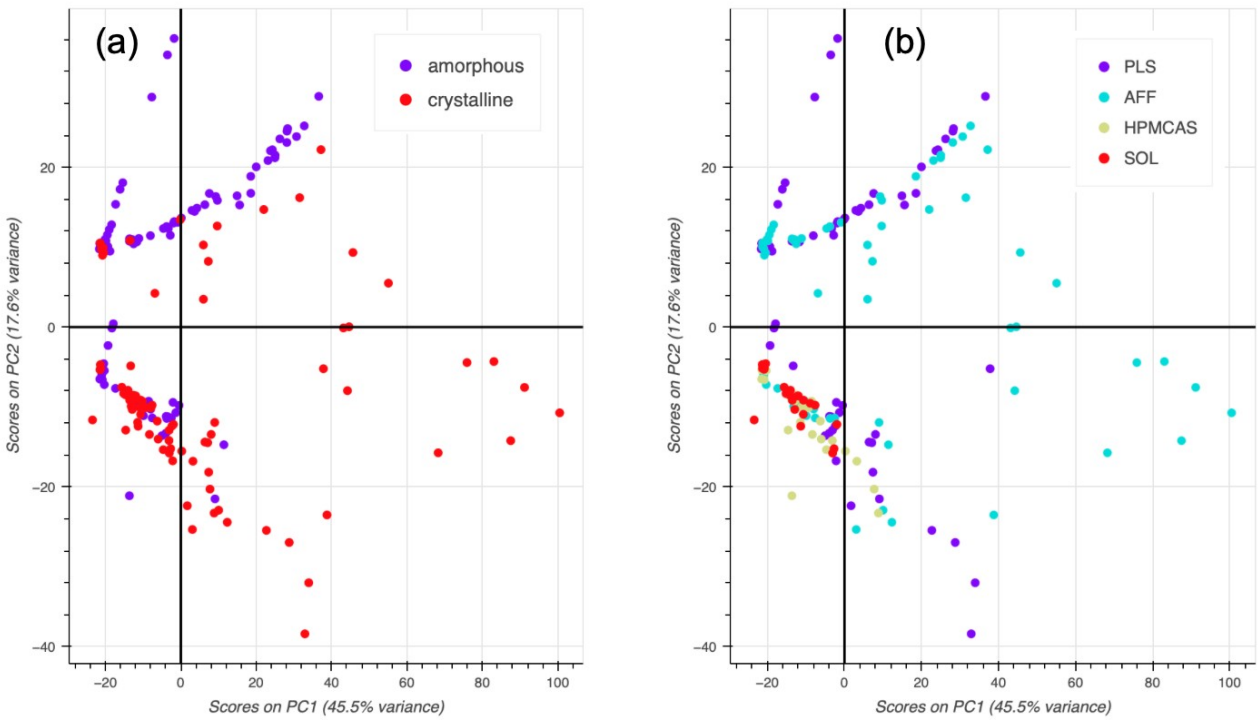

Figure 2. Score scatter plot of PC1 vs PC2 for all samples from the 60 mg/mL and 30 mg/mL plates coloured by **(a)** crystalline vs amorphous and **(b)** polymer. PCA was conducted from 5-2000 cm^{-1}. The variance captured by each principal component is in brackets.

Appendix D

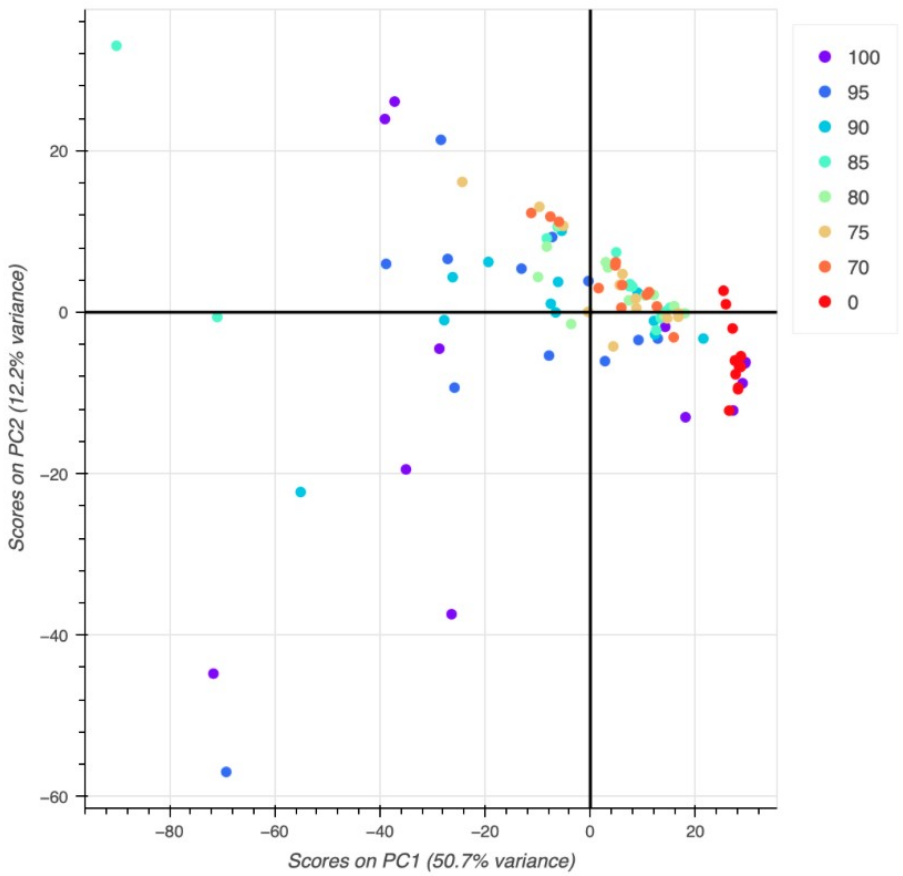

Figure 1. Score scatter plot of PC1 vs PC2 for all samples from the 30 mg/mL plate coloured by drug loading. PCA was conducted from 5-2000 cm^{-1}. The variance captured by each principal component is in brackets.